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Floating marine litter detection algorithms and techniques using optical remote sensing data: A review

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ABSTRACT

Floating Marine Litter (FML) are mainly plastics or synthetic polymers that float on the sea surface after being deliberately discarded or unintentionally lost along beaches, rivers or marine environments. In recent years, much focus has been placed on locating, tracking and removing plastic items in both coastal areas and in the open ocean. The use of high-resolution multispectral satellite images for such purpose is very promising, since satellite images can systematically monitor much larger areas in comparison to the traditional in situ observations. This paper contains a literature review of the published research regarding the optical remote detection of floating marine debris and the proposed associated methodologies. The main aim of this review is to compile all available information on detection methodologies, providing at the same time valuable insights into the different approaches used for floating marine litter monitoring. First, a brief introduction into the theoretical basis of a spaceborne floating marine litter detection system is given. Next, published articles, or relevant research work have been compartmentalised, for analysing the proposed procedures and assisting in a further assessment of their methodological frameworks. Lastly, conclusions and bottlenecks of the existing knowledge on marine litter detection from space are derived. Although the remote detection of floating marine litter is currently limited by inherent restrictions of the available satellite sensors specifications, we highlight how the methodological processing chain can significantly affect the future accuracy of plastic detection from space.

1. Introduction

Marine litter consists of solid materials that have been made or used by people and deliberately discarded or unintentionally lost in in marine and coastal environments, such as wood, metals, glass, rubber, textiles, paper and plastics (UNEP, 2005). Due to its transboundary nature, all the world's oceans and seas, including some of the most remote corners of our planet, now contain substantial amounts of marine litter (Worm et al., 2017). Marine plastics are of particular interest due to the fact that in the last 50 years, plastic production has increased more than 22-fold while the global recycling rate of plastics in 2015 was only estimated at 9% (Geyer et al., 2017). As a result, plastic now accounts for more than 80–90% of all litter found in the marine environment, and it is estimated that in 2016 alone, between 19 and 23 million metric tons of plastics (i.e. 11% of plastic waste generated globally), entered aquatic ecosystems globally, with annual emissions projected at up to 53 million metric tons per year by 2030 under a business-as-usual scenario (Borrelle et al., 2020).

The slow rate of degradation of plastics, combined with a steady increase in the production of single-use items over the last 50 years, has led to a remarkable accumulation of plastic debris at the sea-surface, on the sea-floor and in coastal environments worldwide (Chamas et al., 2020; Smail et al., 2019). This continuous accumulation of plastic has now become a major environmental issue posing a complex multi-dimensional challenge. Plastic debris items in the oceans vary widely in terms of size, shape or chemical composition. Large plastic products and thousands of plastic items can break down into smaller pieces, called microplastics, through a combination of photodegradation, oxidation and mechanical abrasion (Andrady, 2011; Suaria et al., 2016). But in many instances, larger plastic items (i.e. macroplastics) can persist for decades in the ocean especially when shielded from UV radiation under water or in sediments (Gregory and Andrady, 2003), accounting for a large portion of the global mass of floating debris (Lebreton et al., 2019).

The UN Sustainable Development Goals recognized the importance of plastic pollution, establishing a target specifically related to marine litter (SDG target 14.1 and the associated indicator 14.1.1b, "plastic

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debris") and debating the issue in four UN Environment Assembly resolutions (UNEA, 2014, UNEA, 2016, UNEA, 2017, UNEA, 2019). However, large knowledge gaps in terms of understanding marine litter and microplastics still remain: a reliable figure for the volume of plastics entering the ocean, the accumulated volume of plastics in the marine environment, mapping of the source and sink location of plastics, and basic data on the accumulation and transport of floating debris are currently lacking (Zambianchi et al., 2014). There is a need to use and integrate existing data from remote sensing, citizen science, and in situ observations to better understand marine litter dynamics (Maximenko et al., 2019); however, much of the research in this field is at an initial stage and despite the growing interest in floating litter monitoring, a wide range of non-comparable approaches still hampers the development of reliable indicators and spatio-temporal assessments (Galgani et al., 2015).

Macroplastics on the sea surface have been traditionally monitored by ship-based visual surveys, with inherent uncertainty in the data because of the different survey and sampling methods used by different research teams (Suaria and Aliani, 2014). This variability also depends partly on meteorological conditions (Aliani et al., 2003), or the presence over short spatial scales of relevant variability of ocean dynamics, such as coastal currents controlling beach stranding or convergences enhancing offshore accumulation of open ocean macroplastics (Ourmières et al., 2018). In situ estimations can also be performed by numerical models (e.g. Carlson et al., 2017; Eriksen et al., 2014; Mansui et al., 2020; Van Sebille et al., 2015), but more sustained observations are required to unequivocally determine the global marine plastic budget and to support effective management and remediation policies (Thompson et al., 2004).

Satellite remote sensing systems could make a substantial contribution to tackling this problem, although the required technology is still in its infancy (Garaba et al., 2018). Remote sensing satellites are designed to provide observations of global scope, continuous temporal coverage and harmonized data collection and processing, thus potentially being ideal tools for global marine debris monitoring, filling the gaps between sparse in situ observations and providing at the same time, a relatively uniform coverage over larger areas and long time periods (Martínez-Vicente et al., 2019). Nevertheless, although very promising overall, the detection and monitoring of floating marine litter from space is, at least at the current technological level, significantly impacted by physical and technical limitations; such as: atmospheric and sea-surface effects, cloud interference, radiometric, spatial and temporal resolutions of satellite platforms and availability of ground-truthed data.

The current detection methodologies are mainly based on analysis of satellite multispectral imagery. In most cases, the Sentinel-2 satellite is used to detect high concentrations of floating plastics. Satellite hyperspectral imagery is also tested for marine litter detection (Kremezi et al., 2021) and aerial studies showed promising results for the identification of plastics and the discrimination from other floating materials (Garaba et al., 2018; Garaba and Dierssen, 2018, 2019; Balsi et al., 2021). Garaba and Dierssen (2018) examined the spectral response of various plastic objects initially in the laboratory and then using an airborne AVIRIS hyperspectral sensor. They demonstrated very good potential for hyperspectral plastic detection, pinpointing however a significantly reduced response of wet versus dry plastic, indicating a limiting factor regarding detection of floating versus beached litter (Garaba and Dierssen, 2018). Aerial surveys using occupied aircraft present a significant advantage over satellites in terms of spatial resolution, however, such surveys are very limited in terms of acquisition frequency and at the same time exceedingly costly. Hence aerial surveys using full-sized aircraft can mostly be used for proof-of-concept studies and for acquisition of reference data, but cannot be used for large-scale monitoring. Unoccupied aerial systems (UAS) on the other hand are a much less costly alternative to occupied aircraft surveys, but due to technological constraints are still very limited in terms of flight time and payload capabilities, as well as generally requiring a human operator and thus limited in terms of automation potential. Technical advancements will most likely allow for large-scale automated acquisition of remote sensing data using UAS, but to this point satellites are the only platforms capable of continued global coverage.

Global coverage is particularly important due to the fact that low-density plastics can travel over long distances and accumulate over time, although their transport and accumulation dynamics are still largely unknown. Satellites can also help to survey remotely, hard-to-reach areas, in which direct observations are sparse and difficult (Maximenko et al., 2019). At present, however, our ability to address these questions is largely hampered by the evident lack of harmonized approaches and standardized methodologies. The aim of this paper is, to compile all available information on the state-of-the-art marine litter remote sensing detection methodologies, reviewing all different approaches used so-far for floating marine litter monitoring in the marine environment, potentially contributing to the global effort of harmonizing monitoring methods and litter detection protocols.

2. Floating marine litter detection from space: a theoretical perspective

Optical remote sensing classification methodologies are based on the main concept that all pixels of a satellite image that are not of interest need to be disregarded using a specifically formulated procedure. Consequently, a marine litter detection system would use satellite data in order to screen out all non-water pixels, locate pixels containing floating material and successfully discriminate between anthropogenic marine debris and natural phenomena (e.g. foam, white-caps, floating woods, floating macro-algae). Such a system could employ a variety of techniques, from band indices to machine learning, for achieving the goal of floating marine litter detection. Satellite remote sensors are the primary observation tool and need to be carefully selected according to the user requirements in terms of study area or revisit time and, most importantly, the sensor's capabilities regarding spatial resolution, spectral availability and radiometric quality. However, regardless of the data used, the theoretical basis for marine litter detection remains unchanged. Hence, although varying in terms of application method, the main processing steps are usually common in most detection algorithms and can be grouped in two main phases: pre-processing of the data and classification of floating elements into marine debris or "lookalikes" i.e. the phenomena that appear as similar to marine debris.

Pre-processing would include steps such as atmospheric correction, land masking, cloud detection, cloud edge and shadow masking, white caps detection, glint removal and correction. These steps are essential for the removal of atmospheric and sunglint effects and pre-screening of all pixels that can be identified as pixels not containing floating marine litter, like land or cloud pixels and those that do not contain the necessary information to be used in the classification process, e.g. pixels affected by strong glint.

The classification stage would include the processes needed for identification of pixels containing concentrations of FML. This stage would include the identification of pixels that are not screened out through the pre-processing stage, such as vessels and smoke plumes or natural floating materials such as macroalgae and terrestrial vegetation. A number of different methodologies including indices, e.g. the Floating Debris Index (FDI), and machine learning techniques have been proposed to detect FML and the discrimination between natural debris and marine litter. For an FML observation system to be operational in large scales, these processes need to be automated, cover large areas, and work with all available satellite datasets. Verification of the detecting alerts should include ancillary data such as in situ observations of traditional cruises, calibration/validation (cal/val) targets and citizen scientist reports. The development of such classification methodologies requires a significant amount of in situ validated cal/val data. Whether based on indices or machine learning techniques, the spectral information of these datasets is critical in order to achieve the desired

operational accuracy.

3. Review of available research on marine litter detection

This section presents a review of relevant research on the detection of floating marine litter. We have structured the review in a compartmentalised way to maximise the efficiency of the proposed methodologies assessment. This literature review focuses specifically on applications of marine litter detection using multispectral and hyperspectral satellite imagery and does not include the review of research on the spectral properties of marine litter or theoretical approaches. We have included a total of 14 papers, publications and reports, from 2012 to 2021. Fig. 1 presents the main steps that we have used to structure the review of the proposed methodologies with: pre-processing and analysis, classification-processing and results-accuracy assessment. The exact methodologies used (bullet points inside the methodological step boxes) are not exhaustive and mostly act as a guideline for further work, in the process of developing a detection methodology.

Table 1 consolidates the papers and their approach based on our categorisation of information used and main steps. This division is there for better organisation of the present review and does not represent any rigid categorisation, since each approach deals with the issue of FML detection differently.

Novelli and Tarantino (2015) and Acuña-ruz et al. (2018) are two studies that focus specifically on terrestrial rather than marine environments. We believe however that the methods used in these publications have direct or ancillary applications relating to floating marine litter detection and are therefore included in this study. For simplicity the two terrestrial studies are listed in the same table as the marine studies, but are described in a separate section.

Pichel et al. (2012) performed a ghost net marine debris survey in the Gulf of Alaska, using satellite guidance and aircraft observations. The study follows a multi-disciplinary approach, combining remote sensing, modelling and in-situ observations, aiming for detection and ultimately removal of ghost nets from the marine environment. Satellite remote sensing was used mainly for the identification of oceanic features that could be used as proxies for ghost net detection. Identification of eddies was performed through the use of Sea Surface Temperature (SST), chlorophyll and altimeter products, from various (mostly non open access) satellite products. No pre-processing was performed. A combination of Level-3 and Level-4 satellite products were used in a GIS environment to pinpoint and track eddies as a proxy for ghost net detection. Eddie identification was performed manually. The identified and tracked eddy formations were an applicable proxy for ghost net detection.

Aoyama (2016) conducted a study on detection of marine debris in the Sea of Japan using high-spatial-resolution satellite images. A true colour WorldView-2 image was used to arbitrarily identify a suitable area of interest (AOI) for pixel information extraction. The 20 pixels in the AOI were arbitrarily selected for analysis to assess whether they were contaminated with foreign substances. 2D scatter diagrams for all possible WorldView-2 bands combinations for the pixels in the AOI were used to extract suspected marine debris pixels (outlying pixels were deemed as suitable candidates). Spectral Angle Mapper (SAM) was used to select outlying pixels using the mean AOI spectra as the known spectrum and 3 different spectral angles in 4D space (4 bands). Pixels exhibiting higher spectral angles to the mean spectrum were considered candidates for marine debris. The validity of the SAM extraction method was assessed using a low spatial resolution WorldView-2 image of a known target (fixed fishing net with buoys) that could be clearly identified in higher resolution images. No atmospheric correction or other pre-processing was performed. Manual classification of pixels suspected of containing marine debris was performed using scatter diagrams for all possible band combinations and SAM. Accuracy was not specified, but the method appears to successfully classify buoy pixels, although with a large number of false positives. The SAM method successfully identifies outlier pixels in the WorldView-2 image suspected of containing marine debris, with no validation. Validation of the method was performed on a WorldView-2 image known to contain buoys. The buoys were successfully classified but with several false positives.

Topouzelis et al. (2019) examined how plastic litter can be seen in Sentinel-2 images using artificial plastic targets. Three large 10 × 10 m square targets with plastic bags, plastic bottles and fishing net were constructed and deployed at sea. This experiment was the first dedicated study with large artificial plastic targets on the sea surface. High spatial resolution UAV data was collected and an orthophoto map of the study area was produced. Next, direct comparison of the UAV dataset with a Sentinel-2 satellite image was conducted for percentage calculation of plastic coverage in the satellite image. Additionally, spectral behaviour of mixed pixels was performed for identifying the spectral differences

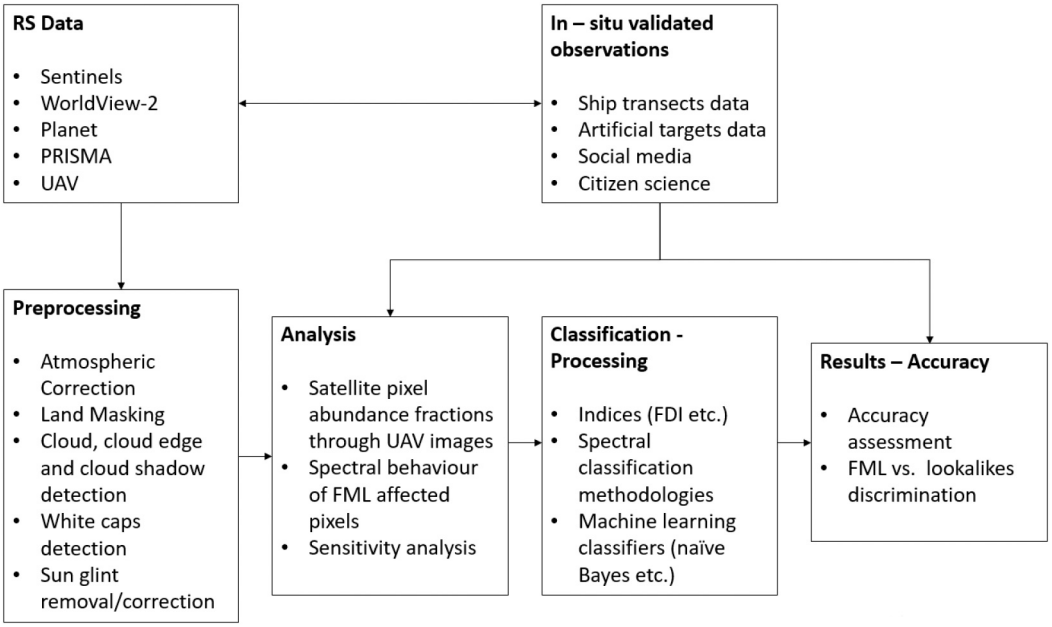

Fig. 1. Flow chart of methodological framework divided into main steps. Exact methodologies used are not exhaustive but present a main guideline for further work.

Table 1
Papers reviewed under the scope of this document.

Paper	RS data	Application	Pre-processing and analysis	Classifiers and processing	Results and accuracy
Pichel et al., 2012	Various oceanographic data	Ghost net detection and retrieval	None, use of ready products	Manual identification and tracking of eddies through altimeter, chlorophyll and SST satellite products	N/A
Novelli and Tarantino, 2015 (terrestrial)	Landsat-8, OLI and TIRS L1	Plastic cover detection, terrestrial	DN to TOA, sun angle correction, no AC, QA bands cloud and corrupt pixel mask, TIRS to at-satellite brightness	4 indices: Tbres (thermal), NDVIgreen, PSI (plastic surface index), NDSI (normalised difference sandy index)	82–89% overall accuracy
Aoyama, 2016	WV-2, WV-3	Floating net detection	identification of spectrally anomalous pixels	Spectral angle mapper	N/A
Acuña-Ruz et al., 2018 (terrestrial)	Lab spectroscopy, WV-3	Beached debris detection	Spectral library creation, Acomp AC of WV3, spectral separability analysis and training of non-parametric methods	Classification and AMD mass prediction	Overall accuracy >75%
Topouzelis et al., 2019	Sentinel-2, Sentinel-1, UAV	Floating ML detection	Sen2Cor and ACOLITE AC, spectral signature extraction	Percentage coverage through UAV and Sentinel-2 data correlation	N/A
Arias et al., 2019	Sentinel-2	Floating ML detection	Sen2Cor AC, cloud and shadow masking, glint correction	Contrast enhancement, red band values, indices (NDVI, NSI, FAI)	Variable with image quality, 50% minimum plastic abundance fraction for identifiable pixel, 57% of used images passed verification criteria
Fronkova, 2019	Sentinel-2	Terrestrial plastic and floating ML detection	Sen2Cor AC	B8-B12 band difference	100% for FML targets, high number of false positives
Biermann et al., 2020	Sentinel-2	Floating ML detection	ACOLITE AC, generation of spectral signatures	FDI and NDVI naïve Bayes probabilistic classification	86% overall for naïve Bayes
Kikaki et al., 2020	Sentinel-2, Landsat-8, Planet	Identification of ML transport and sources	ACOLITE AC, land and cloud masking	None: ML was detected manually through photointerpretation	N/A
Topouzelis et al., 2020	Sentinel-2, UAV	Floating ML detection	ACOLITE AC, generation of spectral signatures, percentage of plastic coverage in S2 pixel through UAV	Reversed linear spectral unmixing, matched filtering	Variable depending on thresholding, high number of false positives
Martinez-Vicente et al., 2020	Sentinel-2 and aerial image	Terrestrial plastic and floating ML detection	Sen2Cor, generation of spectral signatures	FAI, NDVI and NDHI	N/A
Themistocleous et al., 2020	Sentinel-2	Floating ML detection	None	7 well known indices + 2 litter oriented indices: PI (plastic index) and RNDVI (Reversed Normalized Difference Vegetation Index)	N/A
Kremezi et al., 2021	PRISMA	Floating ML detection	Fine coregistration of data, denoising, pansharpening	Hydrocarbon index, Sentinel-2 based index, FDI, 3 novel indices	Detection of plastic targets with few false positives
Basu et al., 2021	Sentinel-2	Floating ML detection	ACOLITE atmospheric correction	2 supervised (support vector regression, semi-supervised fuzzy C-means) and 2 unsupervised (K-means and fuzzy C-means clustering) classification algorithms	Variable based on type of algorithm, over 90% for best performing method

among the plastics and between the surrounding water pixels. In the pre-processing phase, the Sentinel-2 image was atmospherically corrected using Sen2Cor and ACOLITE processors in order to assess the difference in the spectral signature of plastics between the two atmospheric correction algorithms. There was no classification in that study and therefore, no accuracy assessment performed.

In Arias et al. (2019) an Earth Observation (EO) processor for detection of FML using Sentinel-2 data is described. The pre-processing step includes atmospheric correction, cloud screening, sun glint removal, and contrast enhancement. Atmospheric correction is performed using the Sen2Cor algorithm, where L2A products are not available. Cloud screening is performed through the cloud mask provided with the Sentinel-2 product, while detection and removal of cloud shadows is performed with the improved FMask algorithm. Pixels affected by strong glint are excluded from further analysis, while weak glint correction is optionally performed as per Harmel et al. (2018). After pre-processing, the processor classifies remaining surface features in three classes: vegetation, green algae and FML, using a combination of a normalised blue/green index, the NDVI, the NSI (NIR - SWIR index), the floating algae index (FAI) and the Sentinel-2 red band. Results show that in most cases, the processor was capable to mask all non-FML targets, including land and clouds. Several FML events around the world were used to check their effectiveness. The validation process showed that in the case of the artificial floating targets from the Plastic Litter Projects (Topouzelis et al., 2019, 2020), the processor successfully identified the targets on three occasions, while a minimum detection limit was established. Additionally, the processor was able to identify several features that were recognized as windrows of surface material.

Fronkova (2019) assessed a method to map terrestrial and marine plastic litter using open access Sentinel-2 images. Three study sites are used: Eden domes in the UK, the Thilafushi artificial island in the Maldives and the PLP2018 artificial plastic targets in Greece (Topouzelis et al., 2019). For the first study site true colour RGB composites, Normal Difference Vegetation Index (NDVI) and Normalized Difference Built-up Index (NDBI) were used to digitise the Eden Dome plastic surfaces in Sentinel-2 cloud free images. Digitisation of different cover types was performed, and spectral statistics were computed for all cover types. For the PLP targets in Greece, NDWI was used for land masking. In order to distinguish floating plastics from water, the difference in reflectance between bands 8 and 12 was used. Plastic cover of Eden domes showed overall increase in reflectance between bands 5 and 8a with a decrease in SWIR and a peak in band 8. The Thilafushi study area exhibited variation in spectral response compared to the other two study sites, with no reflectance peak in the NIR but in SWIR. The PLP2018 targets

were detected successfully, however, with a large number of false positives (mainly shallow waters, vessels and land-sea interface).

Topouzelis et al. (2020) published a second study on remote sensing of artificial floating plastic targets with data collected in the summer of 2019, during the Plastic Litter Project 2019 (PLP2019). This study was a follow up on the Topouzelis et al. (2019) study. During the PLP 2019 a series of artificial plastic targets were deployed at sea, with a total of 6 Sentinel-2 images being acquired, simultaneously with UAV optical and multispectral data. UAV data are used in combination with Sentinel-2 data in order to calculate the percentage coverage of plastic litter in a Sentinel-2 pixel, through Object Based Image Analysis (OBIA) and image overlay. The pre-processing phase included atmospheric correction using the ACOLITE processor for Sentinel-2 images, and a series of UAV dataset pre-processing i.e. UAV data georeferencing, calculation of percentage coverage and extraction of spectral characteristics of marine litter pixels. A reverse spectral unmixing analysis using the percentage coverage calculated in Topouzelis et al. (2019) is performed in order to obtain the spectral characteristics of the PET targets. The spectral response for PET at Band 8 (NIR) of Sentinel-2 was used to modify the USGS spectral signature for PET (Kokaly et al., 2017). The modified spectral signature is used to perform a matched filtering processing to classify the pixels containing concentrations of plastic litter. Spectral unmixing and matched filtering show promising results. The matched filtering process could be successfully implemented in combination with other methods such as indices in order to assess the concentration of marine litter in a given pixel. Several parameters that affect the classification results are identified, namely: shallows, land-sea interface, vessels, smoke plumes, cloud shadows and unmasked cloud edges, aircraft and glint. Some of these can be singled out through additional processing steps. Depending on the matched filtering thresholds, image quality and overall conditions, marine litter targets can be successfully identified for concentrations of at least as low as 25%. A relatively large number of false positives was classified when applying the process to a whole Sentinel-2 image.

Biermann et al. (2020) conducted a study investigating the detection of floating plastic patches using optical satellite data from the Sentinel-2 satellite and assessed the capability to distinguish those from naturally occurring materials such as seaweed. ACOLITE AC was performed on Sentinel-2 images containing reported and verified floating plastic litter concentrations. NDVI and a novel floating debris index (FDI – modified from the FAI developed for Sargassum detection) (Eq. (1)) were used to identify the floating litter patches.

$$FDI = R_{rs,NIR} - R'_{rs,NIR}$$
$$R'_{rs,NIR} = R_{rs,RE2} + \left(R_{rs,SWIR1} - R_{rs,RE2}\right) \times \frac{(\lambda_{NIR} - \lambda_{RED})}{(\lambda_{SWIR1} - \lambda_{RED})} \times 10 \quad (1)$$

The NDVI was used for discrimination between plastics and naturally occurring materials. Spectral signatures of the identified floating litter from various locations around the world, along with the FDI were used to train a naïve Bayes classifier. Images of plastic targets from PLP 2018 and 2019 were used as validation of the naïve Bayes algorithm. Spectral signatures were generated for a series of detected floating materials and phenomena: seaweed, spume, timber, macroplastics, as well as seawater. Plastic reflectance presented a peak primarily in the NIR part of the spectrum, band 8 (833 nm) of the Sentinel-2 MSI sensor. Seaweed reflectance peaked in the green (560 nm) and red edge (700–780 nm) Sentinel-2 MSI bands, while exhibiting an absorbance feature in the SWIR part of the spectrum, but with a variability which can possibly be attributed to atmospheric correction effects. Timber reflectance peaked in the NIR, with high reflectance also in red and SWIR bands. Pumice was seen as bright across the optical range, reflecting in red, red edge and SWIR and absorbing in NIR. Spume showed the highest reflectance in green and red, with a smaller peak in the NIR band. The naïve Bayes algorithm had an overall accuracy of 86%.

Kikaki et al. (2020) performed a study on remotely sensing the source and transport of marine plastic debris in the Bay Islands of Honduras (Caribbean Sea). The study used high-resolution remote sensing observations for marine litter detection, verified with in situ data, with a purpose to record the spectral characteristics of the litter and identify the source of plastic pollution on the Bay Islands of Honduras. Manual photo-interpretation of floating plastic debris was performed, systematically recording and assessing the spectral signatures of the litter and discriminating from lookalike floating features, e.g. sargassum or foam. Detected areas containing plastics were digitised in three categories: (a) dense, (b) medium and (c) sparse. In total, around 1500 dense, 3200 medium and 5000 sparse pixels were digitised. High-resolution satellite images from multiple sensors (Planet, Sentinel-2, Landsat-8) and situ observations (vessel and diving expeditions) for the period of 2014–2019 were used. The satellite dataset contained 125 Landsat-8 Level1T (June 2014–December 2018) and 340 Sentinel-2 Level1C images. From those 25% were rejected from further analysis due to cloud cover. Also, more than 400 Planet images (with 3 m and 5 m resolution) were observed, 40 of which were further processed for plastic detection and lookalikes discrimination. All satellite data was atmospherically corrected using ACOLITE and surface reflectance values were extracted. The detected plastics were generally observed to accumulate at dense masses following linear trajectories. Around 1600 reflectance values were extracted from the satellite data. The mean observed speed of the litter patches was 6 km d^{-1}. There was no indication of a specific accumulation zone. Finally, it is reported that debris are dispersed in the broader area due to circulation forces.

Martinez-Vicente et al. (2020) investigated the detection of plastics on the shoreline using hyperspectral airbone data and hyperspectral in-situ data and on the sea surface using Sentinel-2 imagery. Preliminary results from hyperspectral reflectance measured from aircraft over targets deployed on a beach have been used to develop Normalised Difference Hydrocarbon Index (NDHI; Aser Mata, pers. comment). This new index, based on the HI - Hydrocarbon Index (Kühn et al., 2004), uses the plastic absorption feature around the 1732 nm band and is designed in a similar fashion to the Normalized Difference Vegetation Index (NDVI). Different plastic types were detected using NDHI. Initial results are promising, showing better results for NDHI than for HI when the entire pixel was covered in plastic. This new index suggests that subpixel detection is possible on dry areas using hyperspectral data. In case pixels are not totally covered by plastics further studies are required to define the minimum coverage percentage. For the sea surface plastics, the study used Sentinel-2 images, atmospherically corrected with the Sen2Cor processor (using the SNAP toolbox). The NDVI was used for detecting aquatic vegetation, in combination with the floating algae index (FAI) for detecting floating objects. The FAI and NDVI combined allowed detection of floating objects and classification in floating vegetation and other phenomena, most probably floating debris.

Themistocleous et al. (2020) investigated the detection of floating plastic litter from space using a Sentinel-2 image and UAS data set (RBG and multispectral at 660 nm and 850 nm) of a single artificial plastic target made from water bottles, sized 3 × 10 m. Spectral signatures of clear water and the plastic target were collected using a portable spectroradiometer from a height of 1.5 and 3 m. The purpose of the study was to examine if Sentinel-2 images can be used to detect plastic litter on the sea surface using spectral indexes. Seven indices were used (i.e. Blue/NIR ratio, NDWI, WRI, NDVI, AWEI, MNDWI, NDMI) and two new were developed: the Plastic Index (PI = B08 / (B08 + B04)) and the Reversed Normalized Difference Vegetation Index (RNDVI = (B04 − B08) / (B04 + B08)). No pre-processing such as atmospheric correction was performed; a Sentinel-2 L1C product was used for the analysis. The study concludes that the plastic litter target was easiest to detect in the NIR wavelengths. The PI index was the optimal index to identify the plastic litter based on a sensitivity analysis. No accuracy analysis was performed. A number of false positives were reported (fishing collars on fish farms, boats with plastic surfaces) when the method was applied in a larger area.

Kremezi et al. (2021) were the first to assess the possibility to employ PRISMA satellite data in order to detect floating marine plastic litter, using artificial marine litter targets. The artificial marine litter targets were constructed and deployed at sea in Mytilene, Lesvos, during a field campaign run in cooperation with the University of the Aegean in the autumn of 2020. Two clear sky PRISMA images are used for the analysis. A variety of pansharpening algorithms are evaluated for their ability to discriminate the plastic target pixels from neighboring water pixels. The pansharpening process aims at increasing the spatial resolution of the hyperspectral PRISMA data by fusing with the panchromatic PRISMA band. A series of required pre-processing steps before the application of the pansharpening algorithms are defined. Level 1 PRISMA data, with no atmospheric correction, are used for the pansharpening process. Fine co-registration between the hyperspectral and panchromatic bands is performed in order to correct for a slight misalignment between the different bands. A novel method for correcting the observed linear periodic noise in the panchromatic images is developed. Five different types of pansharpening methods are applied: component substitution, multi-resolution analysis, hybrid, Bayesian and deep learning. Component substitution pansharpening methods yielded the best results. Principal component analysis, being the simplest method was selected as the most efficient for the study. A series of plastic litter indices were evaluated for use with the pansharpened PRISMA data; more specifically: the hydrocarbon index (Kühn et al., 2004), testing for 19 different band combinations; the Sentinel-2 based index (Papageorgiou, 2019 – diploma thesis) and the floating debris index (Biermann et al., 2020). From these, only the hydrocarbon index using the 866, 951 and 1240 nm bands yielded satisfactory results, although confusing spectral signatures of mixed and PET targets with near-shore water. Three novel indices are proposed for use with the pansharpened PRISMA data. All three indices are simple algebraic functions based on the highest and lowest observed radiances in specific bands, allowing for discrimination between plastic containing and water pixels. The VNIR bands that are used for the three indices are at 492, 596, 719, 781 and 951 nm. A combination of the three indices, using specific thresholds for each one, can satisfactorily detect the floating plastic target material, with a low number of false positives.

Basu et al. (2021) have employed in situ validated Sentinel-2 images of floating marine litter targets, to evaluate the possibility to use supervised and unsupervised classification algorithms to detect the presence of floating marine litter in coastal waters. They used data acquired during experimental campaigns in Greece and Cyprus (Topouzelis et al., 2019, 2020; Themistocleous et al., 2020). The data consisted of five Sentinel-2 images in total. The Sentinel-2 data were divided into calibration (75%) and validation (25%) datasets. The authors used 6 (Blue, Green, Red, RE2, NIR and SWIR1) of the 12 Sentinel 2 bands to perform the analysis, with the 20 m bands being resampled to 10 m. Preprocessing also included atmospheric correction using ACOLITE's dark spectrum fitting atmospheric correction algorithm. Four clustering algorithms were evaluated; two unsupervised – K-means clustering and fuzzy c-means clustering (FCM) and two supervised – support vector regression (SVR) and semi-supervised fuzzy c-means (SFCM). The authors considered three different combinations of bands and indices to develop the attributes for the classification; set A: blue, green, red, RE2, NIR and SWIR1 bands, and the FDI (Biermann et al., 2020) and NDVI indices, set B: red, RE2, NIR and SWIR1 bands, and the FDI and NDVI indices, and set C: FDI and NDVI indices. All clustering methods performed better using the largest attribute set A. The unsupervised algorithms exhibit several misclassifications with false positives and false negatives. The accuracies of the different classification algorithms using attribute set A where: 98.4% for the SVR, 82.2% for the FCM, 81.4% for the K-means clustering, while the SFCM accuracy is reported as being low. In order to perform a sensitivity analysis for the best performing clustering algorithm (SVM using attribute set A), the authors developed a different model using the same number of pixels containing plastic (59 pixels) and of water pixels (118 pixels in total). Overall accuracy and f-score of the SVM algorithm using attribute set A and a total of 118 pixels were found to be 96.7% and 0.97 respectively. The authors conclude that analysis using larger training datasets from different areas is needed for further development of the classification methodologies.

3.1. Terrestrial studies relevant to marine litter detection

Novelli and Tarantino (2015) combined ad hoc spectral indices based on LANDSAT-8 OLI/TIRS sensor data for the detection of plastic covered vineyard. Although this is not a floating marine litter study per se, it is included in the present review on the basis of its potential application to floating marine litter detection. Four Landsat-8 scenes of the area of Taranto, Italy during the summer period were used, utilising both OLI and TIRS sensor data. Four indices were used for classification of plastic cover. The quality assessment bands were used in the sampling process of plastic covered surfaces. OLI digital numbers were linearly converted to top-of-atmosphere reflectance and corrected for sun angle using gains, offsets and local sun elevation angles from the metadata. No atmospheric correction was performed to test the efficiency of the proposed indices apart from atmospheric noise and reduce computation time. The quality assessment bands were also used to mask cloud covered and corrupted pixels. Thermal data were converted to at-satellite brightness temperature using the constants and formula from USGS EROS web site. Pixels were sampled in order to develop the indices to be used for plastic cover identification, through mean reflectance values on different bands. The four normalised indices used where: Tbres: to rescale the brightness temperature domain by standardising the value range with OLI-TOA; NDVIgreen: to discriminate plastic cover from vegetation; PSI: index developed using Tbres and SWIR2 band of L8 for plastic cover detection; NDSI: normalised difference sandy index using SWIR1 and coastal aerosol bands, for sandy surfaces. Classification was performed as an index intersection of the above mentioned four indices using specific thresholds. Final classification resulted in a binary band with plastic surfaces and lookalikes. The use of cloud quality bands reduced the sampling time and improved the collection of spectral information related to plastic surfaces. Overall accuracy was computed through an error matrix and was on average higher than 80%. The method performs well for plastic sheets, which are generally bright in true colour visualisation. It is reported that plastic nets with dense trammel were also detected. Commission errors within urban areas were observed at about 11%.

Acuña-Ruz et al. (2018) performed detection of anthropogenic marine debris over beaches. A novel approach was developed for the identification of macroplastics over beaches with the use of hyperspectral laboratory characterisation and supervised classification methods in high spatial resolution imagery. Laboratory hyperspectral spectroscopy was used to construct a spectral library of collected macrolitter items, determining distinctive spectral bands to be used for classification in satellite imagery. The spectral signatures were convolved using RSRCalculator (a tool to estimate, through numerical convolution, the values corresponding to each spectral range of a given sensor) (Durán-Alarcón et al., 2014). A WorldView-3 scene atmospherically corrected using AComp (Atmospheric Compensation algorithm) was used for classification. The zone in which marine litter was identified (high tide line) was delimited using a digital supervised classification to identify surfaces, vegetation and the intertidal zone. The convolved spectral signatures for the WorldView-3 image were used to train non-parametric methods (support vector machine and random forest) and to perform a linear discriminant analysis. The mass of marine litter was estimated using a stratified sampling method at different distances from the intertidal zone, creating a grid scheme corresponding to the WorldView-3 pixel. An accuracy of 88% was reported for support vector machine classification (highest of all methods applied). Lab spectral analysis showed characteristic absorption features for plastic in the SWIR wavelengths, generally peak reflectance in the NIR and spectral variability evident in the visible range. However, spectral processing

showed inability to classify macrolitter items in separate categories due to spectral similarities and inadequate representation in the WorldView-3 scenes, except for EPS. Overall accuracy between the three classifiers was best for the SVM method.

4. Outlook and discussion

The ever-growing number of studies concentrating on the remote detection of marine litter using satellite data has produced valuable research on a very salient issue. Hence, a series of different methodological approaches have been proposed, based on the data used, the available ancillary information and the particular application.

Most of the selected studies performed atmospheric correction using either the Sen2Cor or ACOLITE processors. Themistocleous et al. (2020) used Sentinel 2 L1C products, while Aoyama (2016) identified spectrally anomalous pixels in WordView data and Pichel et al. (2012) used Level-3 products of oceanographic parameters. While the methodological approaches reviewed above are hard to be grouped in concise categories, the main classification types can be sorted in three groups: i) manual classification (Pichel et al., 2012; Kikaki et al., 2020), ii) use of indices (Novelli and Tarantino, 2015; Arias et al., 2019; Fronkova, 2019; Themistocleous et al., 2020; Biermann et al., 2020; Kremezi et al., 2021) and iii) spectral classification methods or machine learning techniques (Aoyama, 2016; Acuña-Ruz et al., 2018; Topouzelis et al., 2020; Basu et al., 2021).

Several confining factors have been identified in the above-mentioned studies. Basic restrictions are inherent to the specifications of the available satellite sensors, such as the spatial, spectral, radiometric and temporal resolution of a given sensor. Apart from these intrinsic restrictions, a series of essential data processing methodologies can significantly affect the overall capacity for the remote detection of floating marine litter. First and foremost, atmospheric correction of top of atmosphere radiance to bottom of atmosphere reflectance is a necessary pre-processing step that can potentially introduce significant uncertainty in terms of the spectral response of surface features, and hence floating marine litter. For example, ACOLITE and Sen2Cor, two algorithms routinely used for atmospheric correction of Sentinel-2 data, have been shown to produce surface reflectance of significant variability for a given pixel (Topouzelis et al., 2019). Therefore, it is important to assess the capabilities and pertinence of the available atmospheric correction methodologies relating to floating marine litter detection, to confidently reduce the uncertainty inherently present in the atmospheric correction process.

Equally important, the screening out of features and artifacts that can potentially affect the accurate detection of floating marine litter is a process that warrants further investigation. Sunglint for example, is a surface feature prominently present in satellite imagery of marine environments. Sunglint significantly impacts the detection capacity of many remote sensing applications, including the detection of floating marine litter. While strong glint cannot be dealt with and affected pixels need to be excluded from further processing, weak wave glint can potentially be corrected and allow the extraction of meaningful information from the affected pixels. However, most sunglint correction methodologies use the NIR or SWIR bands for estimating and correcting the glint signal contribution. Since these wavelengths are of importance in floating marine litter detection, glint correction inevitably affects the overall performance of detection methodologies. Clouds, cloud edges and shadows, shallow areas, along with other features such as smoke plumes, are also features that can hinder the efficacy of floating marine litter detection. Some of these features can be straightforwardly screened out, however features such as cloud edges can be more difficult to exclude. During pre-processing all the non-relevant phenomena are excluded, allowing the main classifier to separate the plastics from other natural floating materials. Therefore, it can be said that the pre-processing phase is as significant as the actual classifier.

The methodological processing chain, especially the pre-processing step, can significantly affect the future accuracy of plastic detection from space once the processors directly address the atmospheric correction, sunglint, the clouds' effects and accurate land masking. Additionally, future technological improvements in terms of sensor radiometric quality and spatial and spectral resolutions, with allow the use of the parts of the electromagnetic spectrum which are most relevant to floating marine litter detection; i.e. the NIR and SWIR, along with the visible range. Advancements in sensor quality will also allow for better discrimination between floating marine litter and lookalikes, a process which requires high signal-to-noise ratios, since the response of floating materials can easily resemble noise in low signal-to-noise ratio bands and sensors.

Whether implementing simple algebraic approaches such as indices and spectral classification methods, or more complicated classification techniques like artificial intelligence and machine learning, the constraint effects described above remain and need to be accounted for. Perhaps the most important bottleneck however, is the lack of sufficient and high-quality training data for the development of any type of detection algorithm. Real-world scenarios of high-density concentrations of floating marine litter are scarce, subject to availability based on satellite acquisition protocols and highly dependent on weather conditions during acquisition, since even the slightest cloud cover over the area of interest can render any acquired data unusable. Hence, high-quality in-situ validated spectral measurements – along with the corresponding relevant ancillary metadata of floating marine litter are essential, with calibration and validation datasets being an integral requirement for the future development of efficient remote sensing detection methodologies.

The studies presented in this review are essential steps towards achieving accurate, effective and large-scale monitoring of floating marine litter through optical satellite data. Although applicable on their own, further studies incorporating combined solutions, including approaches using non-optical data such as SAR or LIDAR, are most likely needed to overcome the significant physical and technical obstacles to remote sensing of marine litter.

CRediT authorship contribution statement

Konstantinos Topouzelis: Conceptualization, Methodology, Writing, Supervision **Dimitris Papageorgiou**: Data curation, Investigation, Writing-Original draft preparation, Writing-Reviewing and Editing. **Giuseppe Suaria** and **Stefano Aliani**: Investigation, Writing-Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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